Architecturing Crop Models for Intercropping using a Multiscale Modelling Approach

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Process-based crop and plant models

- The challenge in **modelling intercrops**, a cropping system where at least two crop species grow together for a significant portion of their growth cycle (Vandermeer, 1989), lies in the **consideration of interspecific interactions** for resource acquisition (Justes et al., 2021).
- There are two main **process-based modelling approaches**, which are **complementary**, but also separate research communities (Gaudio et al., 2019). *cf Figure A-B*
  - **Crop models** are good at predicting crop yield over several years, consider many plant and soil processes, but do not explicitly consider heterogeneous canopies.
  - **FSPMs** (**F**unctional-**S**tructural **P**lant **M**odels) consider 3D plant structure and compute some processes at organ scale, but imply a high computational complexity.
- **Hybrid approaches** could enable to better account for interspecific interactions in crop models, representing them as emergent properties of the system, simulated through the processes rather than parameters being forced into the crop model (Gaudio et al., 2022).

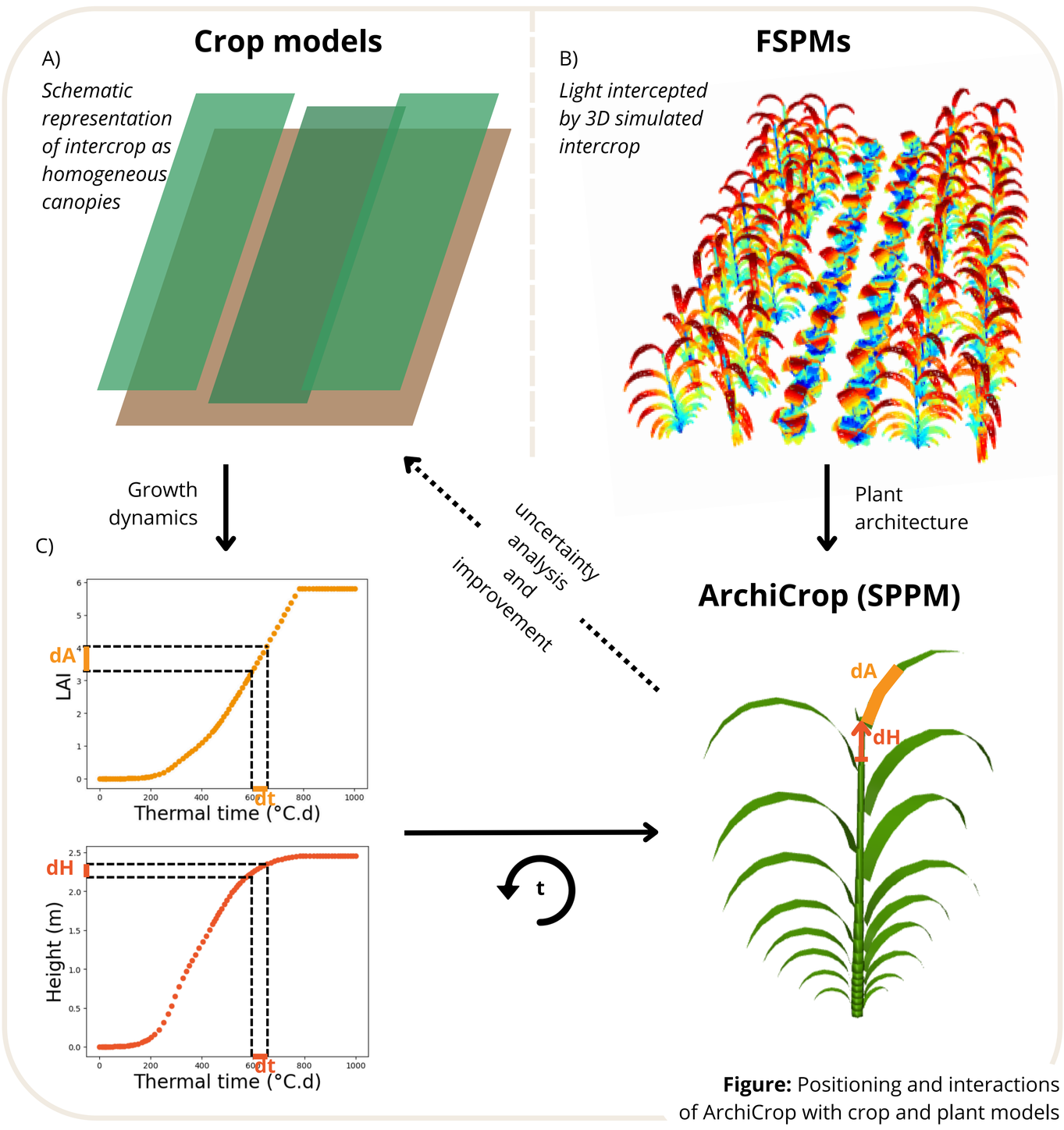

Figure: Positioning and interactions of ArchiCrop with crop and plant models

Key questions

- To what extent could the consideration of **plant architecture** be relevant to improve the **accuracy of crop models** for heterogeneous cropping systems?
- Considering a given growth pattern, what is the impact of **architectural heterogeneity** and **sowing pattern** (spatial and temporal configuration) on **ecophysiological processes**?
- Which plant architectural traits bring most of the **uncertainty** and should be considered explicitly in **crop models**?

Our hybrid multiscale modelling approach

- ArchiCrop model is a **SPPM** (**S**tructural **P**arametric **P**lant **M**odel), i.e. a **3D+t architectured parametric model** that follows the exact **growth trajectory of a crop model**. *cf Figure C*
- Global crop growth is **constrained** by crop model state variables related to canopy growth, e.g. canopy height and LAI (Leaf Area Index).
- The consideration of individual plants and organs in ArchiCrop provides **degrees of freedom** linked to **plant architecture** (i.e. topology, geometry, development) as well as **spatial configuration** for intercrops, introducing **heterogeneity** between plants and organs.
- Growth dynamics already integrate ecophysiological processes, including stresses, computed by the crop model.
- ArchiCrop enables to **visualize in 3D a crop at any stage of growth**, and to see how different plant architectures could correspond to a given growth pattern, showing different hypothetical patterns of structural plasticity.
- ArchiCrop is implemented on OpenAlea platform (Pradal et al., 2008).

OpenAlea

Analysis of crop model hypotheses

- We want to **test crop model hypotheses** for some **ecophysiological processes**, mostly regarding resource acquisition, affected by plant-plant interactions.
- Our method consists in **comparing processes** computed by different models at **different scales** and **analyzing the uncertainty** of key variables related to the processes (e.g. intercepted PAR for light interception), with respect to **architectural traits** and **spatial configuration**.
- *Example of analysis pipeline for light interception process:*
  For a given crop model simulation and for all time steps through the growth cycle: *cf Figure C*
  - **Intercrop ideotype design**: SPPM simulations constrained by growth dynamics from crop model simulation, varying architectural traits and spatial configuration;
  - **Process computation**: compute light interception with a 3D model (e.g. ray tracing model) on the SPPM simulations; *cf Figure B*
  - **Model comparison**: compare the previous results with the intercepted radiation obtained with the crop model (e.g. with Beer Lambert law);
  - **Uncertainty analysis**: on the intercepted radiation to quantify the uncertainty brought by the consideration of architectural traits and spatial configuration.

Conclusion and perspectives

- We propose a new SPPM model parametrized using a **top-down approach** by constraining canopy growth while explicitly considering hypothetical plant architectures. ArchiCrop inherits therefore benefits from FSPM allowing plant insights and structural plasticity studies.
- This hybrid multiscale approach could enable to focus on the architectural traits and spatial configurations that could be relevant to explicitly consider in order to **improve crop models** accuracy, robustness and genericity, while not increasing the number of parameters.
- We will apply this approach to STICS crop model (Vezy et al., 2023), running simulations for sorghum-cowpea intercrop.

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INTERCROP VALUES

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